Practical-Week1

1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

```
ajithmutyala@Ajith:~$ nano process.c
ajithmutyala@Ajith:~$ gcc process.c -o process
ajithmutyala@Ajith:~$ ./process
Enter a Linux command: ls
Parent Process PID: 627
Child Process PID: 628
Executing command: ls
OSSP      assets              hello     process.c
OS_Lab    command_executor.c  process   shell_project
Parent process: Child execution completed.
```

```
ajithmutyala@Ajith:~$ ./process
Enter a Linux command: date
Parent Process PID: 675
Child Process PID: 678
Executing command: date
Fri Aug 14 09:53:31 UTC 2026
Parent process: Child execution completed.
ajithmutyala@Ajith:~$
```

Report: I ran process.c with two commands: ls (parent PID 627, child PID 628) and date (parent PID 675, child PID 678). Both executed correctly with distinct, valid PIDs, and the parent reported completion after each. This confirms the program correctly forks and execs arbitrary user-specified commands.

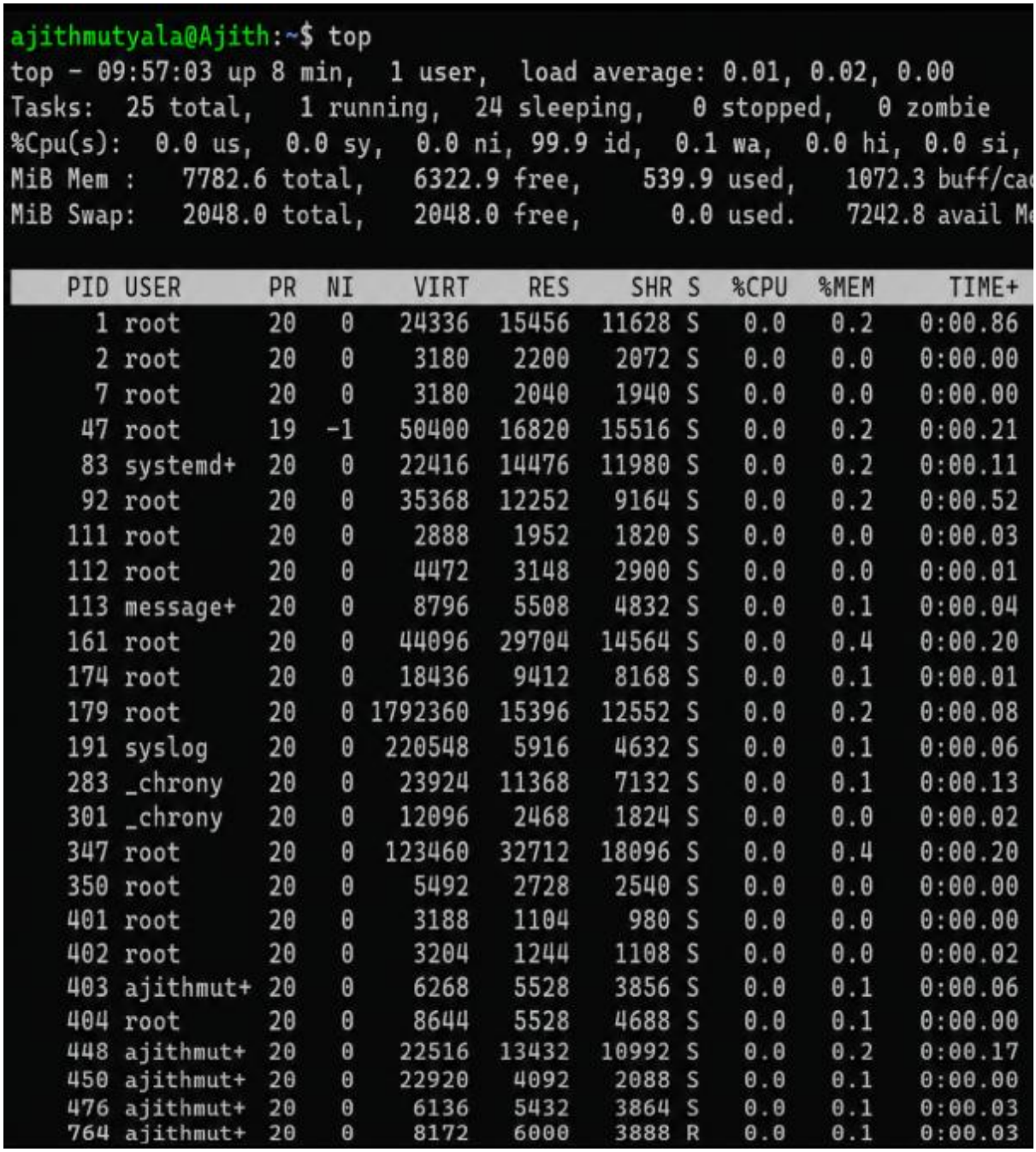
2. Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
ajithmutyala@Ajith:~$ uname -a
Linux Ajith 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun
 18 21:54:43 UTC 2026 x86_64 GNU/Linux
```

```
ajithmutyala@Ajith:~$ lsblk
NAME  MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda     8:0    0 356.9M  1 disk
sdb     8:16   0 159.4M  1 disk
sdc     8:32   0     2G  0 disk [SWAP]
sdd     8:48   0     1T  0 disk /mnt/wslg/distro
```

```
ajithmutyala@Ajith:~$ ps
    PID TTY          TIME CMD
    403 pts/0    00:00:00 bash
    754 pts/0    00:00:00 ps
```

Report: I checked the system using uname -a, lsblk, and ps. uname -a confirmed the kernel (WSL2 Linux). lsblk listed disks including root (sdd, 1T) and swap (sdc, 2G). ps showed the active session's processes: bash (PID 403) and ps (PID 754). This confirms the environment inspection commands correctly reflect the running system.


ajithmutyala@Ajith:~$ top
top - 09:57:03 up 8 min, 1 user, load average: 0.01, 0.02, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.9 id, 0.1 wa, 0.0 hi, 0.0 si,
MiB Mem : 7782.6 total, 6322.9 free, 539.9 used, 1072.3 buff/ca
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7242.8 avail M
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+
1 root 20 0 24336 15456 11628 S 0.0 0.2 0:00.86
2 root 20 0 3180 2200 2072 S 0.0 0.0 0:00.00
7 root 20 0 3180 2040 1940 S 0.0 0.0 0:00.00
47 root 19 -1 50400 16820 15516 S 0.0 0.2 0:00.21
83 systemd+ 20 0 22416 14476 11980 S 0.0 0.2 0:00.11
92 root 20 0 35368 12252 9164 S 0.0 0.2 0:00.52
111 root 20 0 2888 1952 1820 S 0.0 0.0 0:00.03
112 root 20 0 4472 3148 2900 S 0.0 0.0 0:00.01
113 message+ 20 0 8796 5508 4832 S 0.0 0.1 0:00.04
161 root 20 0 44096 29704 14564 S 0.0 0.4 0:00.20
174 root 20 0 18436 9412 8168 S 0.0 0.1 0:00.01
179 root 20 0 1792360 15396 12552 S 0.0 0.2 0:00.08
191 syslog 20 0 220548 5916 4632 S 0.0 0.1 0:00.06
283 _chrony 20 0 23924 11368 7132 S 0.0 0.1 0:00.13
301 _chrony 20 0 12096 2468 1824 S 0.0 0.0 0:00.02
347 root 20 0 123460 32712 18096 S 0.0 0.4 0:00.20
350 root 20 0 5492 2728 2540 S 0.0 0.0 0:00.00
401 root 20 0 3188 1104 980 S 0.0 0.0 0:00.00
402 root 20 0 3204 1244 1108 S 0.0 0.0 0:00.02
403 ajithmut+ 20 0 6268 5528 3856 S 0.0 0.1 0:00.06
404 root 20 0 8644 5528 4688 S 0.0 0.1 0:00.00
448 ajithmut+ 20 0 22516 13432 10992 S 0.0 0.2 0:00.17
450 ajithmut+ 20 0 22920 4092 2088 S 0.0 0.1 0:00.00
476 ajithmut+ 20 0 6136 5432 3864 S 0.0 0.1 0:00.03
764 ajithmut+ 20 0 8172 6000 3888 R 0.0 0.1 0:00.03